

Jiangyan Zhao

Postdoctoral Researcher in Statistics | East China Normal University

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Research Interests

Bayesian nonparametric modeling; Bayesian optimization; clinical trial design; statistical computing and R software.

Education and Research Experience

2024.07-present	Postdoctoral Researcher in Statistics East China Normal University
2017.09-2024.06	Combined Master's and Ph.D. Program in Statistics East China Normal University Research visits at Technion - Israel Institute of Technology: 2021.07-2022.09: Joint Ph.D. visiting student 2020.01-2020.03: Visiting Ph.D. student
2011.09-2015.06	B.Sc. in Mathematics and Applied Mathematics Yantai University

Published and Accepted Papers

BKP: An R Package for Beta Kernel Process Modeling

Jiangyan Zhao, Kunhai Qing, and Jin Xu (2026).

Journal of Statistical Software. Accepted.

<https://arxiv.org/abs/2508.10447>

A Bayesian phase I/II platform design with survival efficacy endpoint for dose optimization

Xian Shi, **Jiangyan Zhao**, Jin Xu, and Rongji Mu (2026).

Statistical Methods in Medical Research, 35(7), 1382-1394. Published.

<https://doi.org/10.1177/09622802261449367>

Bayesian Optimization via Exact Penalty

Jiangyan Zhao and Jin Xu (2024).

Technometrics, 66(3), 368-380. Published.

<https://doi.org/10.1080/00401706.2024.2315937>

Preprints

Shared Keyboard: An improved Bayesian design for phase I clinical trials via Beta kernel process

Jiangyan Zhao, Xian Shi, and Jin Xu (2026).

arXiv:2605.25043.

<https://arxiv.org/abs/2605.25043>

A Unified Framework for Density Estimation under Right-Censored Point-Centred Quarter Sampling

Wenzhe Huang, Guochun Shen, Dingliang Xing, and **Jiangyan Zhao** (2026).

arXiv:2603.08276.

<https://arxiv.org/abs/2603.08276>

Research Software

BKP: Beta Kernel Process Modeling

R software for binomial and multinomial probability modeling, including scalable TwinBKP and TwinDKP implementations.

<https://github.com/Jiangyan-Zhao/BKP>

SKBD: Shared Keyboard Designs

Decision tables, simulations, time-to-event extensions, adaptive dose insertion, and an interactive Shiny application for phase I trials.

<https://github.com/Jiangyan-Zhao/SKBD>

EPBO: Bayesian Optimization via Exact Penalty

Research code for constrained Bayesian optimization using exact penalty functions.

<https://github.com/Jiangyan-Zhao/EPBO>

Teaching Interests

Statistical computing with R; machine learning; optimization methods for statistics; medical statistics.

Academic Profiles

[Google Scholar](#) | [ORCID](#) | [GitHub](#)